Syllabus Details

Syllabus ID:	13166
Syllabus Name:	**Operating System_Hệ điều hành**
Course Name English:	**Operating System**
Subject Code:	**OSG202**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study'
Pre-Requisite:	
Description:	At the end of this course, students will be able to perceive: 1) Background knowledge: (ABET e) -The role of operating system, - important OS concepts, - the mechanism of operating system, and - main problems of Operating system. 2) Practice skills: (ABET k) - Using basic shell command in Linux system fluently. - Fundamental of shell, C language on Linux. - Main problems of Operating system through some simulative exercises. 3) Information searching, reading and selection skills (ABET i) 4) Working in group, documenting and presenation skills (ABET g)
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU LMS https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	- Internet - Linux operating system server and/ or desktop - MARIE simulator - AI tools (ChatGPT/Copilot/Gamma...) and other simulation software if deemed appropriate.
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	932/QĐ-ĐHFPT dated 08/22/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/22/2025

5 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Modern Operating Systems (4th edition or 3rd or 2nd)	Andrew S. Tanenbaum	Pearson	2014 (4th)	4th or 3rd or 2nd	9781292061429 or 9780133591620 (4th Edition)	True	True	False	
2	Connecting with Computer Science	Creg Anderson, David Ferrro, Robert Hilbon	Thomson course Technology	2010	2nd Edition	978-1439080351	False	True	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	Understanding operating systems	Ann McIver McHoes and Ida M. Flynn	Cengage Learning	2008	8th	978-1305674257	False	True	False	
4	http://courses.knox.edu/cs226/						False	False	True	
5	AI Notes:https://www.https://notegpt.io/						False	False	True	

11 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand the main concepts of operating system
2	CLO2	Understand the main concepts of process management
3	CLO3	Understand the main concepts of memory management
4	CLO4	Understand the main concepts of file systems
5	CLO5	Understand the main concepts of I/O systems
6	CLO6	Understand the main concepts of deadlock
7	CLO7	Using Linux operating system
8	CLO8	Using basic shell command in Linux system fluently
9	CLO9	Understand basic Shell language on Linux
10	CLO10	Understand basic C/C++ language on Linux
11	CLO11	Using AI tools to futher explore the main concepts of OS's components

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Chapter 1: Introduction 1.1. What is an operating system	Offline	LO1,LO11	I	Textbook/References Slides ChatGPT()	OSG202	Lecture chapter 1 in the textbook focus on 1.1 Use ChatGPT to explore content more extensively than in the textbook based on keywords provided by the teacher, such as: single user, multi-user, single tasking, multi-tasking	
2	1.2 History of operating system	Offline	LO1,LO11	I	Textbook/References Slides ChatGPT		Lecture chapter 1 in the textbook focus on 1.2 Use ChatGPT to learn more about the history of operating systems based on the keywords provided in class	
3	1.3. Computer hardware review	Offline	LO1	I	Textbook/References Slides		Lecture chapter 1 in the textbook	
4	Chapter 1 Problems	Offline	LO1	ITU	Textbook/References Slides		Lecture chapter 1 in the textbook	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
5	Overview of Linux Operating system Overview of Linux Command	Offline	LO7, LO8,LO11	TU	Tutorial Labs ChatGPT		Research Internet/Textbook/References at home in advance Using ChatGPT to learn basic Linux operating system commands from the textbook Do practice	
6	Short practice of using Linux OS	Offline	LO9,LO11	TU	Tutorial Labs ChatGPT		Research Internet/Textbook/References at home in advance Using ChatGPT to learn basic Linux operating system commands from the textbook Do practice	
7	1.4 The Operating System Zoo 1.5 Operating System Concepts	Offline	LO1,LO11	IT	Textbook/References Slides AI Notes (https://www.https://notegpt.io/)		Lecture chapter 1 in the textbook focus on 1.4, 1.5 Use 'AI Notes' to see additional explanations about different operating systems.	
8	1.6 System Calls 1.7 Operating System Structure	Offline	LO1,LO11	IT	Textbook/References Slides AI Notes (https://www.https://notegpt.io/) ChatGPT		Lecture chapter 1 in the textbook focus on 1.6, 1.7 Use ChatGPT/AI Notes to learn about System Calls and the structures of the Operating System	
9	Chapter 1 Problems	Offline	LO1, LO7	U	Tutorial Labs		Do Lab	
10	Guiding Lab 1	Offline	LO8, LO9	U	Tutorial Labs		Do Lab	
11	Chapter 2: Processes and Threads 2.1 Processes 2.2 Threads	Offline	LO2,LO11	ITU	Textbook/References Slides ChatGPT		Lecture chapter 2 in the textbook focus on 2.1, 2.2 Use ChatGPT to learn about Threads and Processes	
12	2.3 Interprocess Communication	Offline	LO2,LO11	ITU	Textbook/References Slides ChatGPT		Lecture chapter 2 in the textbook focus on 2.3 Use ChatGPT to learn about IPC and the solutions based on the keywords in the lesson	
13	Chapter 2 Problems Review Lab 1	Offline	LO2, LO7, LO8, LO9	U	Tutorial Labs		Do Lab	
14	Guiding Lab 2	Offline	LO7, LO8, LO9	U	Tutorial Labs		Do Lab	
15	Chapter 2 Problems	Offline	LO2, LO7, LO8, LO9	TU	Textbook/References Slides		Lecture chapter 2 in the textbook	
16	2.4 Scheduling	Offline	LO2,LO11	TU	Textbook/References Slides MARIE Simulator ChatGPT/Copilot		Lecture chapter 2 in the textbook focus on 2.4 Use ChatGPT to learn about topics related to FCFS, Shortest Job First, Round Robin, and Priority.	
17	2.5 Classical IPC Problems	Offline	LO2, LO7, LO8, LO9,LO11	TU	Textbook/References Slides ChatGPT		Lecture chapter 2 in the textbook focus on 2.5 Use ChatGPT to learn about topics related to the keywords and solutions in IPC	
18	Chapter 2 Problems	Offline	LO7,LO8, LO9	TU	Textbook/References Slides		Learn Problems	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
19	Shell programming	Offline	LO7, LO8, LO9	T	Tutorial Labs ChatGPT		Do lab Use ChatGPT to learn about commands related to Shell programming	
20	Practice shell programming	Offline	LO7, LO8, LO9	T	Tutorial Labs		Do lab	
21	Practice shell programming (continue)	Offline	LO7, LO8, LO9	U	Tutorial Labs		Do lab	
22	Review Lab 2 Chapter 2 Problems	Offline	LO3, LO8	T	Textbook/References Slides		Learn Problems	
23	Chapter 3: Memory management 3.1 No Memory Abstraction	Offline	LO3, LO8,LO11	T	Textbook/References Slides AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 3 in the textbook focus on 3.1 Use AI Notes Q&A to answer questions related to memory management in the operating system	
24	3.2 A Memory Abstraction: Address Spaces	Offline	LO3, LO8,LO11	T	Textbook/References slides AI Notes (https://www.https://notegpt.io/) Cached simulator		Lecture chapter 3 in the textbook focus on 3.2 Use AI Notes Q&A to answer questions related to memory management in the operating system.	
25	3.3 Virtual Memory	Offline	LO3, LO8,LO11	T	Textbook/References slides AI Notes (https://www.https://notegpt.io/) Cached simulator		Lecture chapter 3 in the textbook focus on 3.3 Use AI Notes Q&A to answer questions related to memory management in the operating system	
26	3.4 Page Replacement Algorithms	Offline	LO3, LO8,LO11	T	Textbook/References Slides ChatGPT/Copilot/Gamma MARIE Simulator		Lecture chapter 3 in the textbook focus on 3.4 Use AI Notes Q&A to answer questions related to memory management in the operating system	
27	Guiding Presentation - topic's selection - Chapter 7: Virtuallization and The Cloud - Chapter 10: CASE STUDY 1: UNIX, LINUX, and ANDROID - Chapter 11: CASE STUDY 2: WINDOWS 8	Offline	LO3, LO8	I	Textbook/References Tutorial Labs Gamma/ AI Notes (https://www.https://notegpt.io/)		Apply the tools that have been learned to develop presentation content	
28	3.5 Design Issues for Paging Systems 3.6 Implementation Issues	Offline	LO3, LO8,LO11	I	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 3 in the textbook focus on 3.5, 3.6 Use AI Notes Q&A to answer questions related to memory management in the operating system	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
29	3.7 Segmentation	Offline	LO3, LO8,LO11	T	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT		Lecture chapter 3 in the textbook focus on 3.7 Use AI Notes Q&A to answer questions related to memory management in the operating system Use ChatGPT to learn more about the mechanism and keywords related to Segmentation and other solutions	
30	Chapter 3 Problems	Offline	LO3, LO8	TU	Textbook/References		Lecture chapter 3 in the textbook	
31	Guilding lab 3	Offline	LO3, LO8	TU	Textbook/References		Lecture chapter 3 in the textbook	
32	Progress Test 1 and Review	Offline	LO3, LO8	TU	Textbook/References		Lecture chapter 3 in the textbook	
33	Review Problems	Offline	LO3, LO8	TU	Tutorial Labs		Do Lab	
34	Chapter 4: Files 4.1 Files 4.2 Directories	Offline	LO4, LO9, LO10	IT	Textbook/References Slides ChatGPT		Lecture chapter 4 in the textbook focus on 4.1, 4.2 Use ChatGPT to learn about the concepts of Files, Directories, and their organization in operating systems.	
35	4.3 File System Implementation 4.4 File System Management and Optimization 4.5 Example File Systems	Offline	LO4, LO9, LO10,LO11	IT	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 4 in the textbook focus on 4.3, 4.4. Use ChatGPT to further explore the contents of the lesson Use ChatGPT or NoteGPT to generate simple file system simulations, then test them in Python or Java use NoteGPT to explore how these file systems worked	
36	Review Lab3	Offline	LO4, LO9, LO10		Tutorial Labs		Do Lab	
37	Guiding Lab 4	Offline	LO4, LO9, LO10	U	Tutorial Labs		Do Lab	
38	Chapter 4 Problems	Offline	LO4, LO9, LO10	U	Textbook/References Slides		Lecture chapter 4 in the textbook	
39	C programming in Linux	Offline	LO4, LO9, LO10	T	Tutorial Labs		Do Lab	
40	Guiding Lab	Offline	LO4, LO9, LO10	T	Tutorial Labs		Do Lab	
41	Chapter 5: Input/Output 5.1 Principles of I/O Hardware 5.2 Principles of I/O Software	Offline	LO4, LO5, LO9, LO10	U	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Do Lab Lecture chapter 5 in the textbook focus on 5.1, 5.2 Use ChatGPT to further explore the contents of the lesson	
42	5.3 I/O Software Layers	Offline	LO5, LO10,LO11	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook focus on 5.3 Use ChatGPT to further explore the contents of the lesson	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
43	5.4 Disks	Offline	LO5, LO10,LO11	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook focus on 5.4 Use ChatGPT to further explore the contents of the lesson	
44	5.5 Clocks 5.6 User Intrefaces: Keyboard, Mouse, Monitor	Offline	LO5, LO10,LO11	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook focus on 5.6 Use ChatGPT to further explore the contents of the lesson	
45	5.5 Clocks 5.6 User Intrefaces: Keyboard, Mouse, Monitor	Offline	LO5, LO10,LO11	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook focus on 5.6 Use ChatGPT to further explore the contents of the lesson	
46	5.7 Thin Clients 5.8 Power Management	Offline	LO5, LO10,LO11	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook fFocus on 5.7, 5.8 Use ChatGPT to further explore the contents of the lesson	
47	Review lab 4	Offline	LO5, LO10	TU	Tutorial Labs		Do Lab	
48	Chapter 5 Problems	Offline	LO5, LO10	TU	Textbook/References Slides		Lecture chapter 5 in the textbook	
49	Chapter 6: Deadlock 6.1 Resources 6.2 Introduction to Deadlocks 6.3 The Ostrich Algorithm	Offline	LO6, LO9, LO10,LO11	IT	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 6 in the textbook focus on 6.1, 6.2, 6.3 Use ChatGPT to further explore the contents of the lesson Use ChatGPT/Copilot to generate scripts that describe deadlock, and then use NoteGPT/ChatGPT/Copilot to explane these scipts.	
50	6.4 Deadlock Detection and Recovery 6.5 Deadlock Avoidance	Offline	LO6, LO9, LO10,LO11		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 6 in the textbook focus on 6.4, 6.5 Use ChatGPT to further explore the contents of the lesson	
51	6.6 Deadlock Prevention 6.7 Other Issues	Offline	LO6, LO9, LO10,LO11	IT	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 6 in the textbook focus on 6.6, 6.7 Use ChatGPT to further explore the contents of the lesson	
52	Chapter 6 Problems Guiding Problems	Offline	LO6, LO9, LO10,LO11	ITU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 6 in the textbook Use ChatGPT to further explore the contents of the lesson	
53	Progress Test 2 and Review	Offline	LO6, LO9, LO10	TU				
54	Review Problems	Offline	LO6, LO9, LO10	TU				
55	Chapter 7, 10, 11 Presentation	Offline						
56	Chapter 7, 10, 11 Presentation	Offline						
57	Chapter 7, 10, 11 Presentation	Offline						

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
58	Chapter 7, 10, 11 Presentation	Offline						
59	Chapter 7, 10, 11 Presentation	Offline						
60	Discussion of current research areas. Review Course	Offline						

0 Constructive question(s)

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Lab	on-going	4	20.0%	>0	in lab session	LO7, LO8, LO9, LO10	practical exercises	N/A	technical aspect of studied topics	guided by instructor	
2	Presentation	on-going	1	20.0%	>0	~20'/group	LO1, LO2, LO3, LO4, LO5, LO6,LO11	use Presentation Rubic sheet	N/A	a selelected topic	prepare at home, working in group, present in class hours	
3	Progress test	on-going	2	20.0%	>0	10'-30'	LO1, LO2, LO3, LO4, LO5, LO6	Multiple choicesMarked by Computer or a suitable format	10-30 (At least 10*n questions where n is the number of Chapters covered in the test.)	Current chapter(s) Each Progess Test covers at least 01 Chapter and at most 03 Chapters, and overall they must cover all learned Chapters of the course.	in class/remote, by instructor	Instruction and shedules for Progress tests must be presented in the Course Implementation Plan approved by director of the campus. Progress test must be taken right after the last lectures of required material. Instructor has resposibility to review the test for students after graded.
4	Final exam	Final exam	1	40.0%	4	60'	LO1, LO2, LO3, LO4, LO5, LO6	Multiple choicesMarked by Computer	50	concepts, algorithms; All chapters from 1 to 6; more than 70% new questions (for the current semester);	by exam board,	